ENGINEERING GEOLOGY

UNIT - III : STRUCTURAL GEOLOGY

Strike, Dip, Outcrops, Folds, Faults, Joints & Unconformities

Syllabus (as given)

- Strike, Dip and Outcrop study of common geological structures associated with rocks such as Folds, Faults, Joints and Unconformities - parts, types, mechanism and their importance in Civil Engineering.

1. Introduction to Structural Geology

Structural Geology deals with the **geometry, origin and attitude of rock structures** produced by deformation (folding, faulting, jointing) of the earth's crust.

- Rocks are deformed by **stress** (compression, tension, shear) generated by plate movements. **Strain** = change in shape / volume. Rock behaviour: **elastic -> ductile (plastic) -> brittle (rupture).**
- **Folds** = ductile deformation; **faults and joints** = brittle deformation.
- Why study? Structures control the **stability of dams, tunnels, slopes, foundations and reservoirs**, and the movement of ground water.

1.1 Attitude of Beds - Strike, Dip and Outcrop

- **Bed (stratum)** = layer of sedimentary rock bounded by bedding planes; **bedding plane** - surface of separation. The orientation of a bed in space is its **attitude**, specified by **strike and dip**.

Strike is 90° to Dip

Strike = trend of a level line
Dip = max. slope of bed
Tool: Brunton / clino compass
e.g. N 40° E / 30° SE

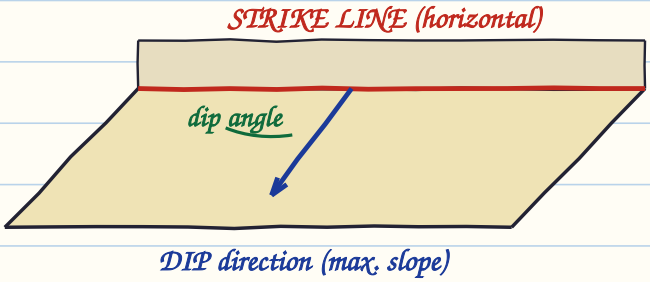


Fig 1: Strike and Dip of an inclined bed

- **Strike:** the direction of the **horizontal line** drawn on the surface of an inclined bed (a line of intersection of bed with a horizontal plane). Given as a **compass bearing**, e.g., N 40° E.
- **True dip:** the **maximum angle of inclination** of the bed with the horizontal, measured **perpendicular to strike**; has both **amount (angle)** and **direction** (e.g., 30° towards SE).
- **Apparent dip:** inclination measured in any direction **other than the true dip direction**; always **less than true dip**.
  - Formula: tan(apparent dip) = tan(true dip) × sin(angle between strike and the section line).
- **Instruments: Brunton / Clar compass** (strike, dip), clinometer.
- Notation on maps: a **long line** shows strike, a **short tick** shows dip direction with the dip angle written beside it. **Horizontal bed** = a circle with a cross (+) inside; **vertical bed** = strike line with a bar across it.

Outcrop

- **Outcrop** = the portion of a rock formation that is **exposed at the earth's surface** (not covered by soil). **Inlier / outlier** are older / younger exposures surrounded by the other.
- **Width of outcrop** depends on: (i) **thickness** of bed, (ii) **dip** of bed, (iii) **topography** (slope of the ground).

Bed attitude	Outcrop pattern on a map
Horizontal bed	Follows **contour lines** (outcrop parallel to contours) - dendritic on dissected terrain
Vertical bed	**Straight line**, not affected by topography; width = thickness
Inclined bed (on plane ground)	Straight belt; width = thickness / sin(dip)

Inclined bed across valley	**"V" shape - Rule of Vs:** V points **in the direction of dip** (beds dipping upstream, or downstream steeper than stream gradient); **exception:** beds dipping downstream at an angle **less than** the stream gradient - V points upstream (against dip)

- **Outcrop width:** steep dip -> **narrow outcrop**; gentle dip -> **wide outcrop**. Strata with the same dip but different thickness show different outcrop widths.
- **Uses of outcrop study:** geological mapping, finding structure (fold/fault), estimating **thickness** and **depth** of beds, locating dam sites and quarries.

True thickness

- $t = w \times \sin(\text{dip})$ on level ground (w = outcrop width). Beds dipping steeply give narrow bands - good for estimating tunnel and slope behaviour.

2. Folds

Fold = a **bend or wave-like curvature of rock layers** formed by plastic deformation due to **compressive stress**, mostly in sedimentary and metamorphic rocks.

2.1 Parts of a Fold

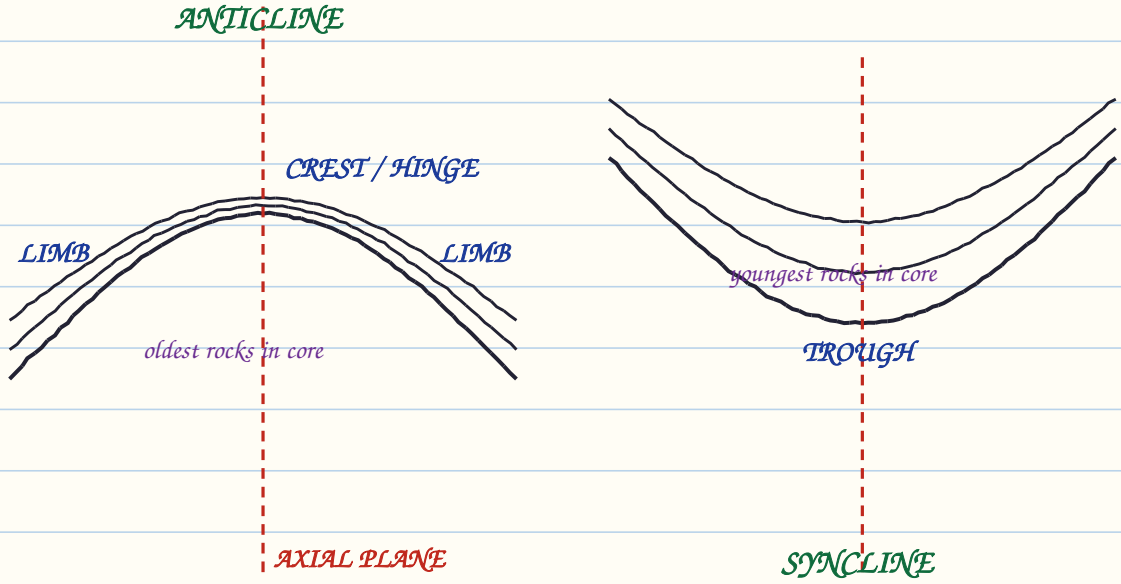


Fig 2: Anticline and syncline with parts

- **Limbs (flanks):** the two sides of a fold. **Hinge (crest/trough line):** line joining points of maximum curvature.
- **Axial plane:** imaginary plane dividing the fold into two halves (as symmetrically as

possible); **axis** = line where the axial plane meets a bed.

- **Crest** (top of anticline), **trough** (lowest part of syncline); **plunge** = inclination of the fold axis with the horizontal; **amplitude** (height), **wavelength**.
- **Inter-limb angle** - angle between the limbs: gentle (> 120°), open, close, tight, isoclinal (0°).

2.2 Types of Folds

Fold	Description
Anticline	Arch-shaped fold: limbs **dip away** from the axis; **oldest beds in the core**
Syncline	Trough-shaped: limbs **dip towards** the axis; **youngest beds in the core**
Monocline	Step-like bend; beds dip in one direction between horizontal strata
Symmetrical	Axial plane vertical, both limbs dip at equal angles
Asymmetrical	Axial plane inclined, limbs dip with unequal angles
Overturned	One limb is **inverted** (both dip in same direction)
Recumbent	Axial plane **horizontal**; fold lies on its side (nappe, Alps / Himalaya)
Isoclinal	Limbs **parallel** to each other, equal dip
Plunging	Axis inclined (not horizontal) - outcrop pattern is V-shaped / horse-shoe
Dome / basin	Beds dip outward in all directions (dome) / inwards (basin) - circular outcrop
Open / close / tight	Based on inter-limb angle

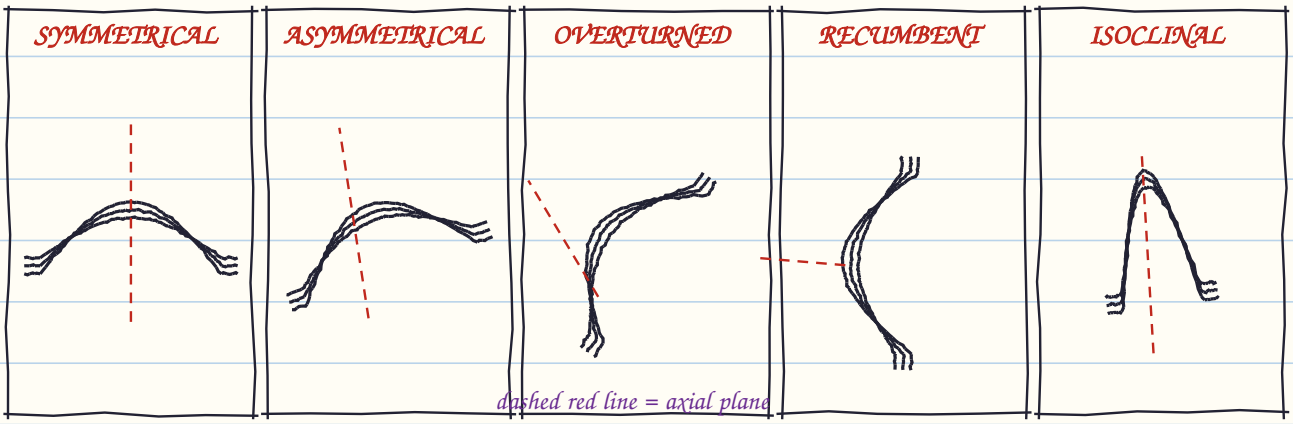


Fig 3: Fold types based on position of the axial plane

2.3 Mechanism of Folding

- **Flexural (buckling) folding:** layers **slip over each other** (like bending a pack of cards) under compression parallel to the beds.
- **Flow folding:** ductile layers flow by plastic deformation - common in rocks at depth, **high temperature**.
- **Shear (slip) folding:** movement along closely spaced shear planes, layers retain thickness measured parallel to axial plane.
- **Drape folding:** over a rising basement block (monocline).
- Controls: type and thickness of rocks, **temperature, confining pressure, rate of strain.**

2.4 Importance of Folds in Civil Engineering

- **Anticline** - tension on the crest -> open **joints, cracks**; suitable for ground water **only in limbs**; unsuitable for **reservoirs** (water leaks along fractures); good for **oil & gas** traps (crest).
- **Syncline** - compression on the trough; **good reservoir sites**; ground water accumulates (**artesian wells** in synclines).
- **Tunnels:** in a **syncline** the core rock is under compression (tight, stable) but **water collects** there - heavy inflow; in an **anticline** the crest has **tension cracks and open joints** (roof falls, over-break) but water drains away. Choose lining and drainage accordingly.
- **Dam sites: dip of beds upstream** (limbs dipping towards the reservoir) is preferred; dip towards downstream = **risk of sliding and seepage**.
- **Slopes:** beds dipping **parallel to slope** and steeper than friction angle -> **slides** (Vajont-type).
- **Highway cuttings:** avoid cutting across the dip direction of weak beds.
- Folded rocks give **trend of strata** - used for locating coal, ore bodies and aquifers.

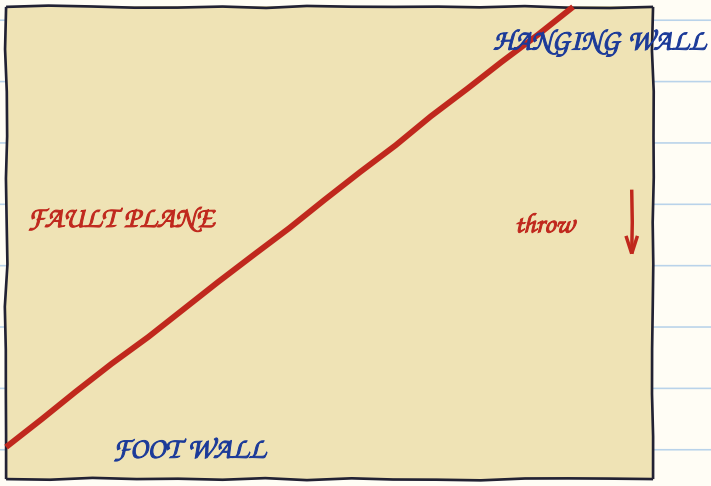
3. Faults

Fault = a **fracture in rocks along which there has been displacement** of the two sides relative to each other. Result of **brittle failure** when stress exceeds rock strength.

- Size: from centimetres to hundreds of km (San Andreas ~1200 km; Narmada-Son

lineament, Godavari rift in India). Movement may be sudden (**earthquake**) or slow (**creep**).

3.1 Parts / Terminology of a Fault



H.W. = block above fault plane
F.W. = block below the plane
THROW = vertical displacement
HEAVE = horizontal displacement
SLIP = true movement on plane
Fault dip = angle of plane

Fig 4: Parts of a fault

- **Fault plane** - surface along which the movement took place; may be vertical, inclined or horizontal; **fault zone** - broken zone with **gouge, breccia, mylonite**.
- **Hanging wall (HW)** - block **above** the inclined fault plane; **Foot wall (FW)** - block **below** it.
- **Dip of fault** - angle between fault plane and horizontal; **Strike of fault** - trend of the fault plane's horizontal line.
- **Throw** - vertical displacement; **Heave** - horizontal displacement; **Slip (net slip)** - actual relative displacement; **Hade** = 90° - dip.
- **Fault scarp** - cliff on the surface; **slickenside** - polished, striated surface; **drag folds** - bending of beds near the fault.

3.2 Classification of Faults

- **By relative movement (most common):**

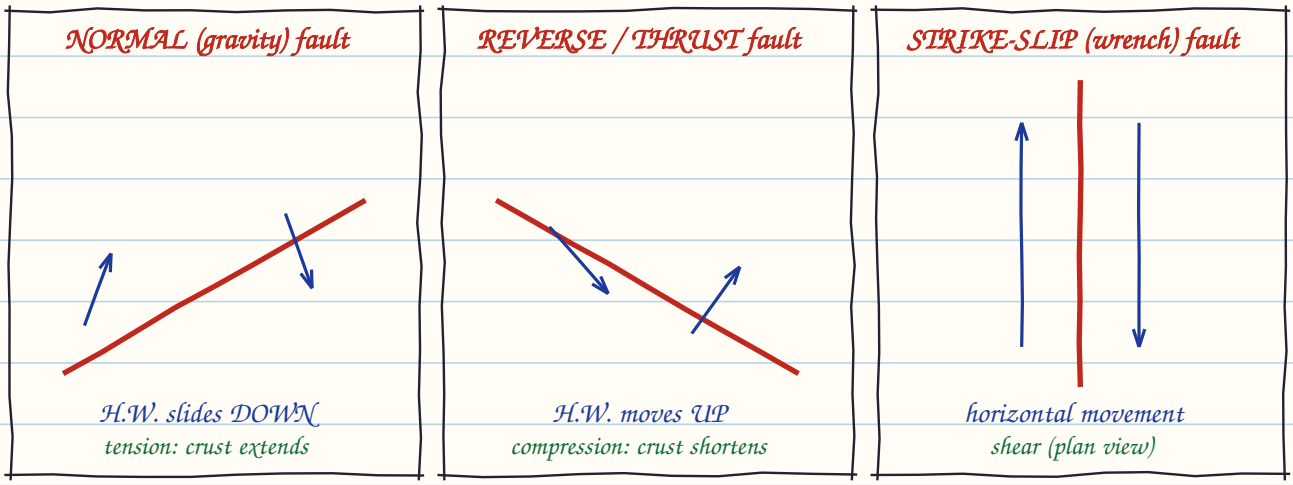


Fig 5: Normal, reverse and strike-slip faults

Type	Features
Normal (gravity) fault	Hanging wall moves DOWN; fault plane steep (60-70°); caused by tension (crustal extension); beds gap on the surface. Forms graben and horst
Reverse fault	Hanging wall moves UP relative to foot wall; caused by compression; beds repeat on the surface; Thrust = reverse fault with low dip (< 45°)
Strike-slip (wrench) fault	Movement horizontal along strike; dextral (right-lateral) or sinistral (left-lateral); vertical plane
Oblique-slip fault	Combination of dip-slip and strike-slip movement
Hinge / rotational fault	One block rotates about a point

- **By angle of dip:** high-angle (> 45°) and low-angle (< 45°).
- **By relation to strike of beds: strike fault** (parallel to strike), **dip fault** (parallel to dip), **oblique fault**.
- **Compound structures: Graben** = down-dropped block between two normal faults (rift valley, e.g., **Godavari valley**); **Horst** = up-thrown block; **step faults** = successive normal faults in same direction; **Nappe / thrust sheet**.

3.3 Causes and Recognition of Faults

- **Causes:** tectonic stress (plate movement), cooling and contraction of earth, **isostatic adjustment**, volcanic activity, gravity sliding.
- **Recognition in the field: sudden break / repetition or omission of beds, fault scarps, slickensides, fault breccia & gouge, drag folds,** offset of streams / ridges, springs and

lines of vegetation, silicification and mineralised zones.

3.4 Importance of Faults in Civil Engineering

- **Dam sites:** faults are **unsuitable** - zone of weak, crushed rock: **leakage, differential settlement**, and possible **reactivation (earthquake)**. Fault is a major seismic risk.
- **Reservoirs:** fault planes act as **channels for water loss**; if reservoir is near an active fault, **induced seismicity** (Koyna).
- **Tunnels:** fault zones give **heavy water inflow, roof falls, squeezing ground** - need special support, grouting; alignment should cross faults at right angles.
- **Foundations & bridges:** avoid active faults; foundation in sheared rock has low bearing capacity.
- **Slopes & roads:** sheared rocks fail; fault zones are weathered deeply.
- **Ground water:** fault gouge may be **impermeable** (barrier) or fault breccia **permeable** (conduit) - springs along faults.
- **Mineral deposits:** faults control ore bodies, oil traps, thermal springs.

Dam & fault - remember

- A dam must NEVER straddle an active fault. If a fault is unavoidable, use flexible design (earth/rock-fill), excavate and treat the gouge (dental concrete, grouting) and design for seismic loads.

4. Joints

Joint = a **fracture or crack in rock with NO appreciable displacement** of the two sides.

Joints are the **most common structure** in rocks.

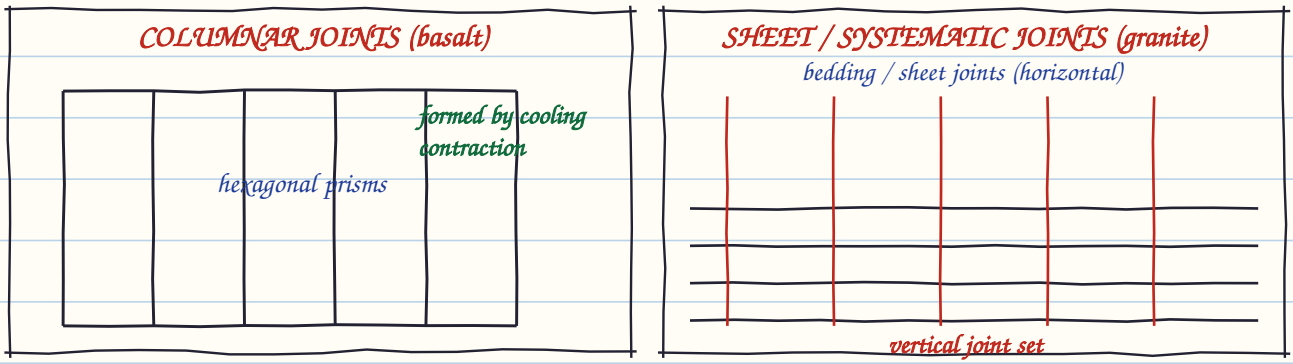


Fig 6: Types of joints

- **Parts:** joint plane, joint set (parallel joints), **joint system** (sets crossing - e.g.,

orthogonal), joint spacing, aperture, filling, persistence.

4.1 Types of Joints

- **Geometric classification: strike joint** (parallel to strike of beds), **dip joint** (parallel to dip), **oblique joint**, **bedding joint** (along bedding plane).
- **Genetic classification:**
  - **Tension joints** - formed by **stretching**, wide, rough, open - cut across rocks.
  - **Shear joints** - formed by shear stress; smooth, closed, occur in **conjugate sets** (about 60° between them).
  - **Columnar joints** - **polygonal (hexagonal) columns** from **contraction on cooling** of lava (basalt, Deccan traps).
  - **Sheet joints (exfoliation)** - curved, parallel to surface, from **unloading / pressure release** (granite domes).
  - **Mural joints** - cubical, in granite from cooling; **Flow joints, cleavage joints, release joints.**

4.2 Importance of Joints in Civil Engineering

- **Strength reduction:** joints are **planes of weakness** - rock mass strength is lower than intact rock strength; define **RQD** (Rock Quality Designation) and rock mass rating.
- **Permeability:** joints are paths for **water leakage** from reservoirs, seepage beneath dams; **grouting** is done to seal joints (curtain grouting).
- **Slope stability: wedge or planar failures** along joints dipping out of slope; rock bolts and anchors needed.
- **Tunnelling: joint orientation** controls roof falls, over-break; best when tunnel axis is **across** the strike of joint sets; **blasting** efficiency.
- **Quarrying:** joints help **block extraction** of dimension stone; closely jointed rock gives small blocks and **poor** building stone.
- **Foundation:** open joints lower bearing capacity; **weathering** proceeds through joints; **ground water** circulation.
- **Aquifers:** jointed granite/basalt forms **hard-rock aquifers** (bore wells in A.P.).

Feature	Fault	Joint	Fold
Nature	Fracture with displacement	Fracture, no displacement	Bend of layers

Stress	Tension / compression / shear	Tension / shear / release	Compression
Effect on dam	Highly dangerous	Needs grouting	Dip direction matters
Remedy	Avoid / treat gouge	Grouting, bolts	Select suitable limb

5. Unconformities

Unconformity = a **surface of erosion or non-deposition** separating younger strata from older rocks - it represents a **time gap (hiatus)** in the geological record. Beds above and below are **not in continuous sequence.**

- Formation steps: older rocks **deposited -> uplifted -> eroded** (or no deposition) -> **sea / land subsides** -> younger beds deposited above the eroded surface.

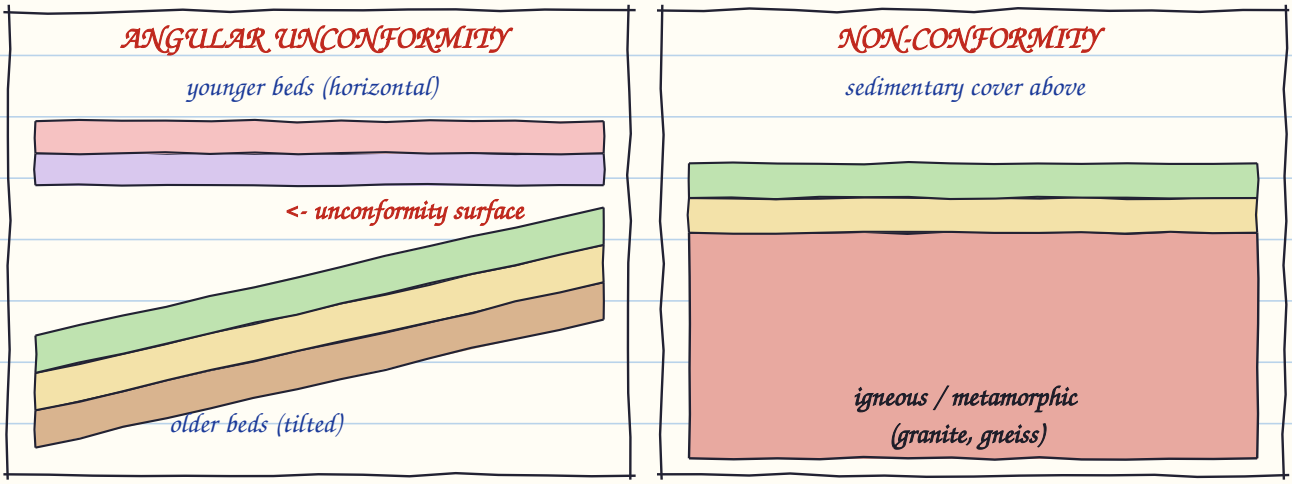


Fig 7: Angular unconformity and non-conformity

5.1 Types of Unconformities

Type	Description
Angular unconformity	Older beds are **tilted / folded**; younger beds above are **horizontal**; angle between them. Shows an earlier **orogeny** (mountain building)
Disconformity	Older and younger beds are **parallel** but a **surface of erosion** (irregular) separates them; evidence: channels, pebbles, soil layer
Non-conformity	**Sedimentary** beds rest on eroded **igneous or metamorphic** rocks (granite, gneiss) - plutonic rocks exposed by erosion

Paraconformity	Parallel beds, **no visible erosion**, only a gap in time (shown by fossils)
Local unconformity	Small-scale, due to local erosion / current action

- **Basal conglomerate** - a layer of pebbles at the base of the younger formation, strong evidence of unconformity.
- **Recognition:** (i) **discordance** of dip and strike, (ii) **basal conglomerate**, (iii) **missing fossil zones**, (iv) **weathered, irregular surface**, (v) abrupt change in rock type or metamorphism, (vi) **mapping** - outcrops of older beds truncated.
- **Overlap / Offlap**, **Outlier / Inlier** are related map patterns.

5.2 Importance of Unconformities in Civil Engineering

- The **unconformity plane** is usually a **weak plane**: weathered, porous and has **high permeability** - leakage in dam abutments and reservoirs; **sliding surface** for slopes.
- Foundations placed across an unconformity face **differential settlement** (hard rock below, soft strata above).
- **Tunnels** crossing an unconformity meet **abrupt change in rock** and possible **water inflow**.
- Indicate **ancient land surfaces** - help in mineral exploration (**bauxite, laterite, placer deposits, iron and manganese** are formed near them) and ground water.
- Basal conglomerate can be a **good aquifer**; also useful marker bed for correlation.

6. Summary - Structures and their Engineering Significance

Structure	Hazard	Engineering action
Anticline	Open joints at crest, leakage	Avoid reservoir on crest; grout
Syncline	Water accumulation	Good reservoir; tunnel drainage
Fault	Weak zone, seismic movement	Avoid / treat / flexible design
Joint	Seepage, rock wedge fall	Grouting, rock bolting, shotcrete
Unconformity	Weak interface, seepage	Excavate to sound rock; cut-off
Dip upstream	Safe	Preferred for dam sites
Dip downstream	Sliding & uplift	Anchorage, drainage, avoid

6.1 Field Measurement Tips (Notes for Lab / Viva)

- Hold the **Brunton compass level** on the bed surface, align edge along the **strike**, read bearing; then measure **dip** with clinometer, perpendicular to strike.
- Always record: **location, rock type, strike, dip amount & direction, thickness, structure** and sketch.
- Quote bearing in **quadrant** (N 30° W) or **azimuth** (330°) form; dip direction by compass quadrant.
- Distinguish **bedding, joint and cleavage** planes before measuring.

6.2 Short Revision Questions

1. **Define strike and dip.** Strike = direction of a horizontal line on the bed; dip = maximum slope angle measured at right angles to strike.
2. **Relation between strike and dip direction?** Always perpendicular (90°).
3. **Name the parts of a fold.** Limbs, crest, trough, axial plane, hinge, axis, plunge.
4. **Difference between anticline and syncline.** Anticline - upward arch, oldest beds in core; syncline - downward trough, youngest in core.
5. **Throw and heave?** Vertical and horizontal displacement across the fault.
6. **Which fault forms under tension?** Normal fault (graben).
7. **What is a basal conglomerate?** Coarse pebble bed at base of a younger formation resting on an unconformity.
8. **Why are faults avoided at dam sites?** Weak sheared zone, leakage and possible reactivation.

Exam tip

- Always draw: (1) strike-dip sketch, (2) anticline/syncline with parts, (3) normal / reverse / strike-slip faults, (4) angular unconformity. Label HW, FW, throw, heave, axial plane.